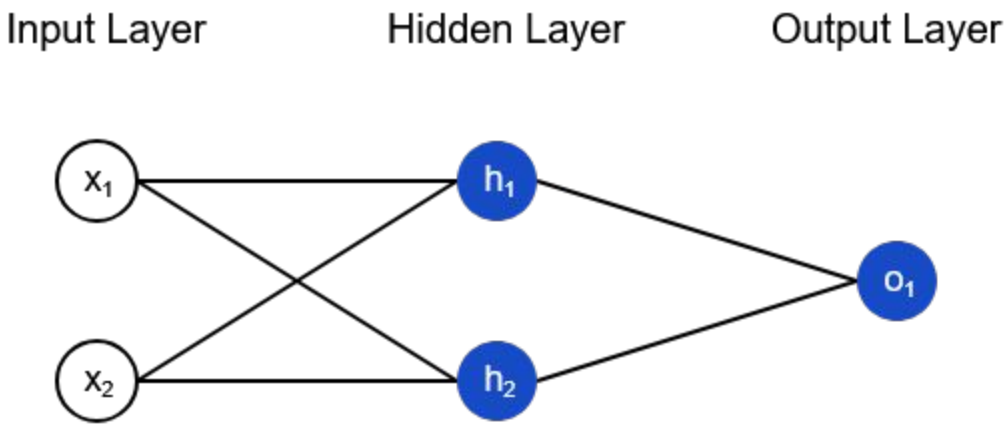
ADAM: A METHOD FOR STOCHASTIC OPTIMIZATION

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ANNs and SGD



Algorithm

1. Iterate *t*
2. Calculate *g_t*
3. Update m_t
4. Update v_t
5. Correct m_t
6. Correct v_t
7. Update parameters

Iterate t

Run a batch of training data through the model.

Requires a set of inputs and labels.

For each input, calculate the loss by comparing the output and the label.

Calculate g_t

For each parameter, calculate the derivative of the loss with respect to that parameter.

Update m_t and correct

$$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$$

$$\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$$

β_1 is some number [0, 1). The paper recommends 0.9.

m_t starts at 0.

Update v_t and correct

$$v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$$
$$\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$$

β_2 is some number [0, 1). The paper recommends 0.999.

v_t starts at 0.

The difference between m_t and v_t is that v_t uses the square of the gradient.

Update parameters

$$\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$$

Theta are the parameters.

Alpha is the learning rate. The paper recommends a value of 0.001.

Epsilon is a small value to avoid division by zero. The paper recommends 10^-8.